EMPLOYEE TABLE

✅ 1. Table Creation Script

```
CREATE TABLE employee (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  gender VARCHAR(10),
  mother_tongue VARCHAR(30),
  department VARCHAR(30),
  salary DECIMAL(10,2),
  dob DATE,
  doj DATE,
  experience INT
);
```

✅ 2. INSERT Script – 50 Employees

```
INSERT INTO employee VALUES
(101,'Arun','Male','Tamil','IT',45000,'1998-05-12','2021-07-01',3),
(102,'Divya','Female','Tamil','HR',38000,'1999-08-20','2022-01-15',2),
(103,'Karthik','Male','Telugu','Finance',52000,'1997-03-10','2020-06-10',4),
(104,'Meena','Female','Tamil','IT',60000,'1996-11-05','2019-08-25',5),
(105,'Rahul','Male','Hindi','Sales',30000,'2000-02-18','2023-02-01',1),
(106,'Sneha','Female','Kannada','HR',42000,'1998-07-22','2021-03-12',3),
(107,'Vijay','Male','Tamil','IT',55000,'1997-09-14','2020-10-05',4),
(108,'Anita','Female','Hindi','Marketing',36000,'1999-12-01','2022-06-18',2),
(109,'Suresh','Male','Telugu','Finance',48000,'1996-01-30','2019-04-10',5),
```

(110,'Pooja','Female','Marathi','HR',39000,'1998-04-09','2021-11-20',3),

(111,'Ramesh','Male','Tamil','Sales',34000,'1997-06-15','2020-02-17',4),

(112,'Kavya','Female','Kannada','IT',62000,'1996-10-11','2018-07-01',6),

(113,'Ajay','Male','Hindi','IT',58000,'1995-12-25','2019-01-10',6),

(114,'Nisha','Female','Tamil','Finance',46000,'1998-03-19','2021-05-05',3),

(115,'Manoj','Male','Malayalam','Operations',40000,'1997-08-02','2020-09-14',4),

(116,'Priya','Female','Tamil','HR',37000,'1999-11-28','2022-08-01',2),

(117,'Deepak','Male','Hindi','Marketing',43000,'1998-02-07','2021-06-21',3),

(118,'Revathi','Female','Tamil','IT',51000,'1997-05-30','2020-01-15',4),

(119,'Sunil','Male','Kannada','Finance',49000,'1996-09-09','2019-12-12',5),

(120,'Aishwarya','Female','Telugu','Sales',35000,'2000-01-18','2023-03-10',1),

(121,'Ganesh','Male','Tamil','Operations',41000,'1997-04-14','2020-07-07',4),

(122,'Shalini','Female','Hindi','HR',36000,'1999-06-21','2022-02-14',2),

(123,'Prakash','Male','Telugu','IT',57000,'1996-12-03','2019-05-01',5),

(124,'Swathi','Female','Kannada','Marketing',39000,'1998-08-16','2021-09-09',3),

(125,'Naveen','Male','Tamil','Finance',53000,'1997-01-27','2020-11-11',4),

(126,'Keerthi','Female','Tamil','IT',60000,'1996-03-05','2018-06-20',6),

(127,'Mohit','Male','Hindi','Sales',32000,'2000-07-08','2023-01-05',1),

(128,'Latha','Female','Tamil','HR',44000,'1997-10-23','2020-04-18',4),

(129,'Sanjay','Male','Marathi','Operations',47000,'1996-02-11','2019-08-30',5),

(130,'Bhavya','Female','Kannada','IT',56000,'1998-12-17','2021-12-01',3),

(131,'Arvind','Male','Hindi','Finance',50000,'1997-09-06','2020-03-25',4),

(132,'Harini','Female','Tamil','Marketing',38000,'1999-05-13','2022-07-07',2),

(133,'Rohit','Male','Hindi','IT',61000,'1996-11-19','2018-09-15',6),

(134,'Yamini','Female','Telugu','HR',40000,'1998-06-01','2021-10-10',3),

(135,'Balaji','Male','Tamil','Sales',36000,'1999-01-09','2022-04-04',2),

(136,'Neha','Female','Hindi','Finance',48000,'1997-07-27','2020-02-20',4),

(137,'Kiran','Male','Kannada','Operations',42000,'1998-03-16','2021-06-06',3),

```
(138,'Sandhya','Female','Tamil','IT',59000,'1996-08-08','2019-01-01',5),
(139,'Lokesh','Male','Telugu','Marketing',45000,'1997-11-04','2020-12-12',4),
(140,'Anusha','Female','Telugu','HR',37000,'1999-02-25','2022-09-09',2),
(141,'Vimal','Male','Tamil','IT',54000,'1997-04-03','2020-05-15',4),
(142,'Preethi','Female','Tamil','Finance',41000,'1998-09-29','2021-08-08',3),
(143,'Sathish','Male','Tamil','Operations',43000,'1996-06-18','2019-03-03',5),
(144,'Ritu','Female','Hindi','Marketing',39000,'1999-10-10','2022-11-11',2),
(145,'Madhan','Male','Tamil','Sales',35000,'2000-12-01','2023-06-01',1),
(146,'Pallavi','Female','Marathi','HR',42000,'1997-02-14','2020-07-20',4),
(147,'Harsha','Male','Telugu','IT',58000,'1996-05-26','2018-12-12',6),
(148,'Uma','Female','Tamil','Operations',40000,'1998-01-05','2021-04-04',3),
(149,'Nitin','Male','Hindi','Finance',51000,'1997-08-31','2020-09-09',4),
(150,'Saranya','Female','Tamil','IT',62000,'1996-04-22','2018-01-10',6);
```

#PART A – Basic SELECT & Filtering (10 Questions)

#1. Display all records from the employee table.

```
SELECT * FROM EMPLOYEE;
```

#2. Show only employee name, department, and salary.

```
SELECT EMP_NAME, DEPARTMENT, SALARY
FROM EMPLOYEE;
```

#3. List all employees working in the IT department.

```
SELECT * FROM EMPLOYEE
WHERE DEPARTMENT = 'IT';
```

#4. Display employees whose salary is greater than 50,000.

```
SELECT *
```

```
FROM EMPLOYEE
WHERE SALARY > 50000;
```

#5. List all female employees.

```
SELECT * FROM EMPLOYEE
WHERE GENDER = 'FEMALE';
```

#6. Display employees whose mother tongue is Tamil.

```
SELECT * FROM EMPLOYEE
WHERE MOTHER_TONGUE = 'TAMIL';
```

#7. Show employees who joined the company after 2021-01-01.

```
SELECT * FROM EMPLOYEE
WHERE DOJ > '2021-01-01'
ORDER BY DOJ ASC;
```

#8. Display employees whose experience is less than 3 years.

```
SELECT * FROM EMPLOYEE
WHERE EXPERIENCE < 3;
```

#9. List employees belonging to HR or Finance departments.

```
SELECT * FROM EMPLOYEE
WHERE DEPARTMENT = 'HR' OR 'FINANCE';
```

#10. Display employees whose salary is between 40,000 and 60,000.

```
SELECT * FROM EMPLOYEE
WHERE SALARY BETWEEN 40000 AND 60000;
```

#PART B – Sorting, DISTINCT & Pattern Matching (10 Questions)

#11. Display employees ordered by salary in descending order.

```
SELECT EMP_ID, EMP_NAME, DEPARTMENT, SALARY
FROM EMPLOYEE
ORDER BY SALARY DESC;
```

#12. List employees ordered by date of joining (DOJ).

```
SELECT * FROM EMPLOYEE
ORDER BY DOJ;
```

#13. Display distinct departments in the company.

```
SELECT DISTINCT DEPARTMENT
FROM EMPLOYEE;
```

#14. Display distinct mother tongues of employees.

```
SELECT DISTINCT MOTHER_TONGUE
FROM EMPLOYEE;
```

#15. List employee names that start with ‘A’.

```
SELECT EMP_NAME
FROM EMPLOYEE
WHERE EMP_NAME LIKE 'A%';
```

#16. List employee names that end with ‘a’.

```
SELECT EMP_NAME
FROM EMPLOYEE
WHERE EMP_NAME LIKE '%a';
```

#17. Display employees whose name contains the letter ‘i’

```
SELECT * FROM EMPLOYEE
WHERE EMP_NAME REGEXP 'i';
```

#18. Display employees sorted by department and salary.

```
SELECT * FROM EMPLOYEE
ORDER BY DEPARTMENT, SALARY;
```

#19. List employees ordered by experience (highest first).

```
SELECT * FROM EMPLOYEE
ORDER BY EXPERIENCE DESC;
```

#20. Display the top 5 highest paid employees.

```
SELECT * FROM EMPLOYEE
ORDER BY SALARY DESC
LIMIT 5;
```

#PART C – Aggregate Functions (10 Questions)

#21. Find the total number of employees

```
SELECT COUNT(EMP_NAME)
FROM EMPLOYEE;
```

#22. Find the average salary of all employees

```
SELECT AVG(SALARY) AS AVG_SALARY
FROM EMPLOYEE;
```

#23. Find the highest salary in the company

```
SELECT MAX(SALARY) AS HIGHEST_SALARY
```

```
FROM EMPLOYEE;
```

#24. Find the lowest salary in the company.

```
SELECT MIN(SALARY) AS MINIMUM_SALARY
FROM EMPLOYEE;
```

#25. Count the number of female employees.

```
SELECT COUNT(GENDER) AS GEN_FEMALE
FROM EMPLOYEE
WHERE GENDER = 'FEMALE';
```

#26. Find the average salary of male employees.

```
SELECT AVG(SALARY) AS MALE_AVG_SALARY
FROM EMPLOYEE
WHERE GENDER = 'MALE';
```

#27. Count the number of employees in the IT department.

```
SELECT COUNT(*) AS IT_EMP
FROM EMPLOYEE
WHERE DEPARTMENT = 'IT';
```

#28. Find the total salary paid to all employees.

```
SELECT SUM(SALARY) AS TOT_SAL
FROM EMPLOYEE;
```

#29. Find the average experience of employees.

```
SELECT AVG(EXPERIENCE) AS AVG_EXP
FROM EMPLOYEE;
```

#30. Count employees whose salary is more than 45,000.

```
SELECT COUNT(*) FROM EMPLOYEE
WHERE SALARY > 45000;
```

#PART D – GROUP BY & HAVING (Single Table) (10 Questions)

#31. Display department-wise employee count

```
SELECT DEPARTMENT, COUNT(*)
FROM EMPLOYEE
GROUP BY DEPARTMENT;
```

#32. Find department-wise average salary.

```
SELECT DEPARTMENT, AVG(SALARY)
FROM EMPLOYEE
GROUP BY DEPARTMENT;
```

#33. Display mother tongue-wise employee count.

```
SELECT MOTHER_TONGUE, COUNT(*)
FROM EMPLOYEE
GROUP BY MOTHER_TONGUE;
```

#34. Display gender-wise employee count.

```
SELECT GENDER, COUNT(*)
FROM EMPLOYEE
GROUP BY GENDER;
```

#35. Find departments having more than 5 employees.

```
SELECT DEPARTMENT, COUNT(*) FROM EMPLOYEE
```

```
GROUP BY DEPARTMENT
HAVING COUNT(*) > 5;
```

#36. Display mother tongues having average salary above 45,000.

```
SELECT MOTHER_TONGUE, AVG(SALARY)
FROM EMPLOYEE
GROUP BY MOTHER_TONGUE
HAVING AVG(SALARY) > 45000;
```

#37. Find departments where maximum salary exceeds 60,000.

```
SELECT DEPARTMENT, MAX(SALARY) FROM EMPLOYEE
GROUP BY DEPARTMENT
HAVING MAX(SALARY) > 60000;
```

#38. Display departments with average experience greater than 4 years.

```
SELECT DEPARTMENT, AVG(EXPERIENCE)
FROM EMPLOYEE
GROUP BY DEPARTMENT
HAVING AVG(EXPERIENCE);
```

#39. Find gender-wise average salary.

```
SELECT GENDER, AVG(SALARY)
FROM EMPLOYEE
GROUP BY GENDER
HAVING AVG(SALARY);
```

#40. Display departments having total salary above 3,00,000.

```
SELECT DEPARTMENT, SUM(SALARY)
```

```
FROM EMPLOYEE
GROUP BY DEPARTMENT
HAVING SUM(SALARY) > 300000;
```

🔹 PART E – Date & Experience-Based Queries (10 Questions)

#41. Display employees born after 1998.

```
SELECT * FROM EMPLOYEE
WHERE DOB > '1998-12-31'
ORDER BY DOB ASC;
```

#42. Find employees who joined in the year 2020.

```
SELECT * FROM EMPLOYEE
WHERE YEAR(DOJ) = 2020 ;
```

#43. Display employees whose date of birth month is May.

```
SELECT * FROM EMPLOYEE
WHERE MONTH(DOB) = 3 ;
```

#44. Find employees with more than 5 years of experience.

```
SELECT * FROM EMPLOYEE
WHERE EXPERIENCE > 5;
```

#45. Display employees who joined in the last 3 years.

```
SELECT *
FROM EMPLOYEE
WHERE YEAR(DOJ) >= (
   SELECT MAX(YEAR(DOJ)) - 2  #If you want N calendar years including the latest year:latest_year - (N - 1) -> N = 3 -> 3 − 1 = 2
```

```
  FROM EMPLOYEE
)
ORDER BY DOJ DESC;
```

#46. Find the youngest employee (based on DOB).

```
SELECT *
FROM EMPLOYEE
ORDER BY DOB DESC
LIMIT 1;
```

#47. Find the most experienced employee.

```
SELECT *
FROM EMPLOYEE
WHERE EXPERIENCE = (SELECT MAX(EXPERIENCE) FROM EMPLOYEE);
```

#48. Display employees whose age is greater than 25 years.

```
SELECT EMP_ID, EMP_NAME, DEPARTMENT, DOB
FROM EMPLOYEE
WHERE TIMESTAMPDIFF(YEAR, DOB, CURDATE()) > 25;
```

#49. Find employees who joined before 2020.

```
SELECT * FROM EMPLOYEE
WHERE DOJ  < '2020-01-01';
```

#50. Display employees who joined in the same year as their experience value (logic-based).

```
SELECT *
FROM EMPLOYEE
```

```
WHERE EXPERIENCE = (YEAR(CURDATE()) - YEAR(DOJ));
```

#Bonus / Challenge Questions (Optional)

#51. Categorize employees as Low / Medium / High salary using CASE WHEN.

```
SELECT EMP_ID,
    EMP_NAME,
    SALARY,
    CASE
      WHEN SALARY < 40000 THEN 'LOW'
      WHEN SALARY BETWEEN 40000 AND 50000 THEN 'MEDIUM'
      ELSE 'HIGH'
    END AS SALARYCATEGORY
FROM EMPLOYEE;
```

#52. Calculate current age of each employee using DOB.

```
SELECT * , TIMESTAMPDIFF(YEAR, DOB, CURDATE()) AS AGE
FROM EMPLOYEE;
```

#53. Calculate experience from DOJ instead of using the experience column.

```
SELECT EMP_ID,
    EMP_NAME,
    DOJ,
    TIMESTAMPDIFF(YEAR, DOJ, CURDATE()) AS EXPERIENCE
FROM EMPLOYEE;
```

#54. Display employees whose salary is above department average (can be attempted later with subqueries).

```
SELECT EMP_ID,
```

```
    EMP_NAME,
    DEPARTMENT,
    SALARY
FROM EMPLOYEE
WHERE SALARY > (
  SELECT AVG(SALARY)
  FROM EMPLOYEE
);
```

#55. Identify employees eligible for promotion (experience ≥ 5 and salary ≥ 50,000).

```
SELECT EMP_ID,
    EMP_NAME,
    DOJ,
    SALARY,
    TIMESTAMPDIFF(YEAR, DOJ, CURDATE()) AS Experience
FROM Employee
WHERE TIMESTAMPDIFF(YEAR, DOJ, CURDATE()) >= 5
 AND Salary >= 50000;
```